

Privacy-Preserving Machine Learning with Federated Learning and Differential Privacy: A Comprehensive Academic Survey

Abstract

Federated Learning (FL) has fundamentally redefined distributed machine learning by enabling collaborative model training across decentralized edge devices and institutional silos without centralizing raw data. Despite adhering to the principle of data minimization, FL remains highly susceptible to sophisticated privacy threats, including gradient inversion, membership inference, and property inference attacks. Consequently, Differential Privacy (DP) has emerged as the definitive mathematical framework for rigorously quantifying and mitigating privacy leakage in FL systems. This comprehensive survey provides a systematic examination of Differentially Private Federated Learning (DP-FL). We structurally analyze the theoretical foundations of DP-FL, including Rényi Differential Privacy (RDP) and the Moments Accountant, which enable tight privacy composition over iterative training. The manuscript systematically categorizes DP-FL into Central DP, Local DP, Distributed DP with Secure Aggregation, and Shuffle DP, evaluating the unique privacy-utility-computation trade-offs of each architecture. Furthermore, we explore cutting-edge algorithmic optimizations for managing non-IID data heterogeneity, adaptive gradient clipping, and communication sparsification. Crucially, the survey investigates the integration of DP into complex paradigms, including Vertical Federated Learning (VFL) and the parameter-efficient fine-tuning (PEFT) of Large Language Models (LLMs) via Low-Rank Adaptation (LoRA). Finally, we address an emerging paradigm shift regarding the tension between DP and Byzantine robustness, illuminating how privacy noise can inadvertently mask sophisticated backdoor attacks. This survey consolidates the theoretical and empirical state-of-the-art, charting the trajectory for future privacy-preserving distributed intelligence.

Introduction

The unprecedented efficacy of modern deep learning, spanning high-dimensional computer vision to vast generative language models, is predicated on the availability of massive, diverse datasets. Historically, the prevailing paradigm required centralizing this data into a monolithic repository for stochastic gradient descent (SGD) optimization¹. However, as machine learning permeates sensitive domains—such as healthcare diagnostics, financial forecasting, and mobile user modeling—centralized data collection has encountered insurmountable legal, ethical, and logistical barriers. Regulatory frameworks, most notably the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA), mandate strict

data governance, while corporate entities increasingly guard proprietary data as core intellectual property¹.

Federated Learning (FL) was introduced by McMahan et al. as a decentralized alternative that brings the code to the data¹. In the standard FL protocol (e.g., Federated Averaging, or FedAvg), a central server initializes a global model and broadcasts it to a cohort of participating clients. Each client executes multiple steps of local optimization using its private dataset and transmits only the resulting model update (or gradient) back to the server. The server aggregates these updates to refine the global model, iterating this process until convergence⁵. By ensuring that raw data never leaves the client device, FL intrinsically supports data minimization and physical isolation⁵.

Despite its decentralized nature, FL does not guarantee formal privacy⁸. A robust body of adversarial machine learning literature has demonstrated that intermediate model parameters and gradients encode highly structured representations of the underlying training data⁹. Operating under an "honest-but-curious" threat model, an adversarial server or malicious co-client can exploit these updates. Membership Inference Attacks (MIAs) can reliably determine whether a specific data record was part of a client's training set⁸. More alarmingly, Gradient Leakage Attacks (GLAs) and Gradient Inversion techniques—such as Deep Leakage from Gradients (DLG) and GradInversion—cast privacy breaching as an optimization problem¹⁰. By iteratively perturbing dummy inputs to minimize the distance between their generated gradients and the intercepted client gradients, adversaries can achieve pixel-perfect reconstructions of training images and token-for-token recoveries of text sequences¹⁴. To systematically defend against these vulnerabilities, Differential Privacy (DP) is integrated into the FL lifecycle¹⁸. Differential privacy provides a mathematically rigorous, unconditional upper bound on the privacy loss incurred by releasing the output of a statistical query or algorithm²¹.

Formally, a randomized algorithm $\mathcal{M}$ satisfies (ϵ, δ) -differential privacy if, for any two adjacent datasets D and D' differing by at most one record, and for any subset of possible outputs S , the following inequality holds²¹:

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta$$

The privacy budget ϵ quantifies the maximum acceptable privacy loss, while δ represents the probability of catastrophic privacy failure. The integration of DP and FL—yielding Differentially

Private Federated Learning (DP-FL)—is achieved by bounding the L_2 -sensitivity of the model updates via gradient clipping, and subsequently injecting calibrated statistical noise (e.g., Gaussian or Laplace noise) into the updates¹⁰.

However, applying DP in federated environments introduces a multidimensional optimization challenge. Injecting sufficient noise to satisfy stringent privacy budgets inevitably corrupts the semantic integrity of the gradients, leading to utility degradation, slower convergence, and

heightened sensitivity to data heterogeneity (non-IID distributions)¹⁷. Furthermore, the architectural placement of the noise injection—whether on the client side, the server side, or distributed among cryptographic primitives—defines fundamentally different trust assumptions and communication overheads⁵.

This survey provides a comprehensive review of the DP-FL landscape. The remainder of the paper is structured as follows. The *Methods* section systematically categorizes the research area, beginning with the theoretical foundations of privacy accounting, followed by detailed subsections on Central DP, Local DP, Distributed DP (with Secure Aggregation), and Shuffle DP. It then expands into domain-specific applications, including Vertical Federated Learning (VFL) and Large Language Models (LLMs). The *Challenges and Prospects* section evaluates critical unresolved issues, focusing heavily on the paradoxical relationship between DP and Byzantine robustness, as well as strategies for addressing non-IID client drift. Finally, the *Conclusion* summarizes the trajectory of the field.

Methods

The integration of differential privacy into federated learning is not a monolithic technique but rather a spectrum of methodologies defined by their threat models, cryptographic augmentations, and aggregation mechanisms. This section dissects the fundamental building blocks and distinct architectural paradigms of DP-FL.

1. Theoretical Foundations: Granularity and Privacy Accounting

Before evaluating specific architectures, it is necessary to establish the theoretical mechanisms that govern privacy accounting in iterative machine learning protocols.

1.1 Record-Level vs. User-Level Differential Privacy

In the context of machine learning, the definition of "adjacent datasets" D and D' dictates the granularity of protection. Standard DP-SGD, as utilized in centralized learning, provides *record-level* (or item-level) DP, ensuring that the model output is indistinguishable whether a single data sample is included or not³.

In FL, however, a single client typically contributes a local dataset comprising hundreds or thousands of records (e.g., a user's entire messaging history or photo gallery). Protecting only a single record is often insufficient. Consequently, DP-FL literature frequently targets *user-level* DP, introduced by McMahan and Geyer⁴. User-level DP is a strictly stronger guarantee requiring that the global model remains indistinguishable whether an entire client (and all their associated records) participates in the training or completely drops out⁴. Achieving user-level DP requires bounding the sensitivity of the entire aggregated client update, which heavily penalizes clients with large or highly informative datasets, necessitating advanced sparsification and regularization techniques to prevent severe accuracy degradation²⁶.

1.2 Moments Accountant and Rényi Differential Privacy (RDP)

Federated training is an iterative process requiring hundreds of communication rounds. Under basic composition theorems, the privacy budget ϵ degrades linearly with the number of

rounds, leading to intolerably fast budget depletion. To enable deep learning with DP, Abadi et al. introduced the Moments Accountant, which tracks the log-moment generating function of the privacy-loss random variable across iterations¹⁹.

This framework was mathematically formalized and generalized via Rényi Differential Privacy (RDP) by Mironov³⁴. A randomized mechanism $\mathcal{M}$ satisfies (α, ϵ) -RDP if, for all adjacent datasets D, D' , the Rényi divergence of order α is bounded by ϵ :

$$D_\alpha(\mathcal{M}(D) \parallel \mathcal{M}(D')) \leq \epsilon$$

RDP offers a quantitatively tighter and operationally convenient method for tracking cumulative privacy loss across sequential FL rounds. It easily converts back to standard (ϵ, δ) -DP at the end of training. Crucially, RDP seamlessly models *privacy amplification by subsampling*—a core characteristic of FL where only a random fraction of the total client population is sampled in each communication round¹⁹. By tracking the moments across subsampled Gaussian mechanisms, RDP allows FL models to train for thousands of rounds while maintaining single-digit ϵ values¹⁹.

2. Central Differential Privacy (CDP) and Adaptive Clipping

Central Differential Privacy (CDP) assumes the existence of a fully trusted central server. In this paradigm, clients compute their local updates on their private data and transmit plaintext updates to the server. The server computes the aggregated update, applies a gradient clipping threshold, and injects Gaussian noise prior to updating the global model⁵.

The seminal algorithm in this space is DP-FedAvg. To bound the L_2 -sensitivity of the aggregation, each client's update Δ_k is clipped to a predefined maximum threshold C . If the update norm exceeds C , it is scaled down: $\tilde{\Delta}_k = \Delta_k \cdot \min(1, C/\|\Delta_k\|_2)$ ²¹. The server then computes the noisy average:

$$\Delta = \frac{1}{|K|} \left(\sum_{k \in K} \tilde{\Delta}_k + \mathcal{N}(0, \sigma^2 C^2 I) \right)$$

2.1 The Challenge of Fixed Clipping Thresholds

In CDP, the selection of the clipping threshold C creates a severe bias-variance trade-off³⁸. An excessively large C forces the server to inject massive amounts of noise (since noise variance scales with C^2), drowning out the signal. Conversely, a small C introduces heavy structural bias by truncating legitimate gradient directions, particularly in non-IID settings where local

updates vary wildly⁴⁰. Furthermore, as the global model converges, the average magnitude of client updates naturally decays; a fixed C chosen at the start of training will inevitably become too large in later rounds, leading to noise domination³⁸.

2.2 Adaptive Clipping Mechanisms

To resolve this, recent literature has proposed Adaptive Clipping algorithms. The DP-LAC (Lightweight Adaptive Clipping) algorithm dynamically adjusts C during training without consuming additional privacy budgets for hyperparameter tuning³⁸. DP-LAC utilizes private histogram estimation to deduce the optimal threshold based on the actual distribution of update norms at a given round. Other quantile-driven approaches estimate the global clipping threshold by taking the median or a specific quantile (e.g., 75th percentile) of the incoming update norms, simultaneously allowing clients to adjust their local noise scales⁴¹. Component-by-component clipping strategies have also been proposed to handle differing variation ranges across distinct layers of deep neural networks, significantly mitigating signal loss compared to flat vector-level clipping⁴⁰.

3. Local Differential Privacy (LDP) and Sparsification

While CDP provides the highest theoretical utility, it relies on an honest curator. In real-world deployments (e.g., cross-device FL on millions of smartphones), the server is often untrusted. Local Differential Privacy (LDP) addresses this by shifting the noise injection directly to the client side. Each client applies a randomized mechanism $\mathcal{M}_{LDP}$ to their update before it leaves the device, ensuring that the server only ever receives cryptographically deniable data⁵.

3.1 The Curse of Dimensionality in LDP

The fundamental limitation of LDP in deep learning is the curse of dimensionality⁴⁵. To satisfy (ϵ, δ) -LDP, a client must add noise proportional to the dimensionality d of the model. In modern models with millions of parameters, the localized noise variance becomes astronomical. When the server aggregates thousands of these highly noisy vectors, the variance compounds, often preventing model convergence entirely¹³.

3.2 Compression, Quantization, and Sparsification

To make LDP viable, researchers employ aggressive dimensionality reduction. Techniques such as CAFL (Compressed Adaptive FL) utilize Compressive Sensing (CS) to drastically shrink the size of the local model updates before applying LDP noise⁴⁵. By reducing the parameter count by up to 95%, the requisite noise scale is proportionally minimized, yielding zero-bias estimators with vastly improved signal-to-noise ratios⁴⁵.

Similarly, random sparsification methodologies (e.g., Fed-SPA) mandate that clients transmit only the top- k most significant gradient coordinates, masking the rest as zero. This "sparsification-amplified privacy" restricts the maximum influence of any data point to a tiny subset of parameters, structurally reducing sensitivity and allowing for lower noise

magnitudes²⁶. Stochastic quantization techniques, such as QMGeo, replace additive continuous noise with probabilistic rounding protocols based on Mixed Truncated Geometric Distributions, providing formal LDP guarantees solely through the randomness of the quantization process, thereby eliminating the need for independent noise injection⁴⁹.

4. Distributed Differential Privacy (DDP) and Secure Aggregation

Distributed Differential Privacy (DDP) seeks the "best of both worlds": the superior model utility of CDP combined with the robust trust model of LDP⁴⁷. DDP achieves this synergy by leveraging cryptographic Secure Multi-Party Computation (SMPC), specifically Secure Aggregation (SecAgg)²⁰.

4.1 The SecAgg Protocol

In DDP, clients are not required to independently satisfy full LDP. Instead, each client adds only a small fraction of the total required Gaussian noise to its local update⁵². Before transmission, clients engage in a SecAgg protocol (typically based on additive secret sharing and pairwise key exchanges) to encrypt their updates⁷. The server aggregates these encrypted payloads. Due to the properties of modular arithmetic over finite groups, the pairwise cryptographic masks mathematically cancel out during summation, allowing the server to decrypt only the aggregate update⁷. Because the server is cryptographically blinded to the individual updates, the privacy guarantee is analyzed over the sum of the clients' noise components, effectively simulating a trusted central curator⁵².

4.2 The Distributed Discrete Gaussian Mechanism

A profound technical barrier to combining DP with SecAgg is the datatype mismatch: standard DP mechanisms rely on continuous real numbers (floating-point Gaussian/Laplace noise), while

cryptographic secret sharing operates strictly over discrete finite fields (integers modulo q)²⁴. Standard floating-point arithmetic is vulnerable to finite-precision attacks, where adversaries exploit rounding errors to reverse-engineer exact inputs²⁴.

To solve this, Kairouz et al. formalized the Distributed Discrete Gaussian Mechanism⁵². By stochastically rounding the scaled gradients to an integer grid and drawing noise from a

mathematically pure Discrete Gaussian distribution $\mathcal{N}_{\mathbb{Z}}(\mu, \sigma^2)$, the protocol supports exact modular arithmetic. This ensures that cryptographic masks perfectly cancel and the exact DP bounds are preserved, achieving CDP-level accuracy using less than 16 bits of precision per coordinate²⁴.

4.3 Reducing SecAgg Overhead

The primary drawback of SecAgg is its massive communication and computational cost, which

scales $\mathcal{O}(n^2)$ with the number of clients due to pairwise interactions⁷. Recent innovations utilize sparse Johnson-Lindenstrauss (JL) random projections to compress updates prior to

SecAgg, reducing the communication cost from $\tilde{\mathcal{O}}(d)$ to $\tilde{\mathcal{O}}(\min(n^2\epsilon^2, d))$ bits per

client²⁰. These dimensional reduction strategies have made DDP feasible for large-scale, cross-device edge computing networks⁶¹.

5. The Shuffle Model of Differential Privacy

The Shuffle Model of DP has recently emerged as an intermediate architecture that minimizes trust assumptions without requiring the heavy cryptographic overhead of SMPC⁴⁷. It introduces a semi-trusted third party—the "shuffler"—situated between the clients and the aggregation server⁶³.

5.1 Privacy Amplification by Shuffling

In Shuffle DP, clients apply a relatively weak local randomizer (satisfying ϵ_0 -LDP) to their updates, encrypt them using the server's public key, and transmit them to the shuffler. The shuffler randomly permutes the order of the incoming messages, completely decoupling the data from the originating client identities, and forwards the anonymized multiset to the server⁶⁶.

The theoretical breakthrough of this model is *privacy amplification by shuffling*. As analyzed by Balle, Feldman, and Erlingsson, the act of hiding an individual's noisy report within a crowd of n other noisy reports exponentially increases the adversary's uncertainty⁶⁶. If n clients utilize ϵ_0 -LDP, the resulting central privacy guarantee ϵ_c perceived by the server is approximately⁶³:

$$\epsilon_c = \mathcal{O} \left(\log \left(1 + \frac{(e^{\epsilon_0} - 1)^2}{n} \right) \right) \approx \mathcal{O} \left(\sqrt{\frac{e^{\epsilon_0} \log(1/\delta)}{n}} \right)$$

This amplification factor of $\mathcal{O}(n^{-1/2})$ is transformative: clients can inject dramatically less noise locally, maintaining high signal fidelity, while the system still achieves strict central DP guarantees⁶³.

5.2 Blanket Divergence and Concurrent Shufflers

Advanced theoretical work on the "Privacy Blanket" bounds the exact information leakage under finite n scenarios using blanket divergence metrics⁶⁶. Furthermore, "concurrent shufflers" have been proposed for streaming or continuous FL scenarios. By routing overlapping batches of user data through multiple asynchronous shufflers of varying sizes, the protocol achieves optimal error bounds of $\tilde{\mathcal{O}}(n^{1/(2k+1)})$ where k is the number of shufflers, vastly outperforming sequential, single-batch shuffle models⁷².

6. Privacy in Vertical Federated Learning (VFL)

The standard FL frameworks discussed thus far assume Horizontal FL (HFL), where clients share identical feature spaces but possess distinct data samples. Vertical Federated Learning

(VFL), by contrast, addresses cross-silo collaborations where multiple institutions share the same user base but hold disparate, proprietary feature sets (e.g., an e-commerce platform holding purchasing history and a bank holding credit scores for the same customers)⁷⁴.

6.1 Vulnerabilities: Embedding and Label Leakage

In a typical VFL training iteration (split learning), clients with features ("bottom models") compute forward embeddings and transmit them to the active client holding the target labels ("top model"). The active client calculates the loss and backpropagates the gradients to the passive clients².

This architecture is vulnerable to two distinct attacks: *Embedding Leakage*, where an adversary reconstructs raw input features from the transmitted forward representations, and *Label Leakage*, where passive clients infer highly sensitive proprietary labels from the backpropagated gradients³. Furthermore, relation inference attacks can deduce private graph links (e.g., social connections) from intermediate activations⁷⁵.

6.2 DP Mechanisms in VFL

Applying DP to VFL is uniquely challenging because noise must be injected into intermediate activations rather than explicit model parameters⁷⁴. To mitigate label leakage, Label Differential Privacy directly perturbs the gradient of the loss function at the active client. Advanced defenses incorporate distance correlation penalties, explicitly optimizing the bottom models to minimize the statistical dependence between the transmitted embeddings and the actual labels, stunting the adversary's ability to extract label intelligence⁷⁷.

In highly sensitive collaborations, DP is combined with Additively Homomorphic Encryption (HE) or secret sharing (e.g., the FedVS framework)². By encrypting the embeddings, the top model can compute aggregated sums and backpropagate errors purely in the ciphertext domain, guaranteeing that neither colluding passive clients nor the active client can inspect the plaintext representations⁷⁴.

7. Large Language Models (LLMs) and Parameter-Efficient Fine-Tuning

The contemporary frontier of DP-FL involves the collaborative fine-tuning of Large Language Models (LLMs) across distributed datasets. Standard full-parameter DP-FL is fundamentally incompatible with LLMs⁷⁹. DP theory dictates that the variance of the injected noise must scale

linearly with the parameter dimensionality d . Injecting noise across 100 billion parameters utterly obliterates the semantic signal, rendering the model untrainable¹¹.

7.1 The LoRA Gradient Coupling Problem

To circumvent this, DP-FL is applied exclusively via Parameter-Efficient Fine-Tuning (PEFT) algorithms, predominantly Low-Rank Adaptation (LoRA)⁸³. LoRA freezes the massive

pre-trained weights and introduces two small trainable matrices, A and B , into each

transformer layer, where the update is represented as $\Delta W = A \times B$ ⁵⁰.

However, direct application of DP to LoRA induces a critical failure mode: *Gradient Coupling*

and Noise Amplification⁷⁹. If isotropic DP noise is added to both matrix A and matrix B independently during local training, their subsequent multiplication during the forward pass yields a cross-term of the noises. This quadratic noise amplification drastically degrades the performance of large vision and language models⁷⁹.

7.2 Architectures for DP-LoRA

To resolve the gradient coupling dilemma, several specialized DP-LoRA architectures have emerged:

- **FFA-LoRA (Federated Freeze-A LoRA):** This protocol initializes matrix A randomly and permanently freezes it across all clients. Only the zero-initialized matrix B is fine-tuned and perturbed with DP noise. By restricting the noise to a single matrix, the quadratic amplification is eliminated, and communication costs are halved⁵⁰.
- **FedASK (Double Sketching):** Freezing A limits the expressiveness of the model. FedASK uses randomized Singular Value Decomposition (SVD) to project updates into a shared low-rank subspace. It securely aggregates the sketched updates and reconstructs both A and B dynamically on the server, allowing full representational capacity without the penalty of compounding local noise⁷⁹.
- **LA-LoRA (Local Alternating LoRA):** This algorithm temporally decouples the matrices, updating matrix A in one communication round and matrix B in the next. This sequential alignment isolates the DP noise, restoring SOTA accuracy on benchmarks like RoBERTa and Swin Transformers under strict privacy budgets ($\epsilon = 1$)⁸³.

Table 1: Comparison of DP-PEFT Architectures for LLMs

Framework	Core Mechanism	Noise Amplification Handling	Expressiveness	Communication Cost
Standard DP-LoRA	Add noise to A and B	High (Quadratic compounding)	Full	Standard LoRA
FFA-LoRA 50	Freeze A , Train B only	Eliminated	Restricted	50% Reduction
FedASK 82	SVD Double Sketching	Low (Global reconstruction)	Full	Standard LoRA

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LA-LoRA 83	Alternating round updates	Eliminated (Temporal isolation)	Full	Standard LoRA

Challenges and Prospects

While the integration of DP into FL has matured theoretically, severe practical challenges persist, primarily concerning system security and non-IID data convergence.

1. The Paradox of DP and Byzantine Robustness

A deeply entrenched assumption in early FL literature was that Differential Privacy inherently provides Byzantine robustness against malicious actors⁸⁷. Byzantine adversaries seek to execute poisoning attacks or implant backdoors, manipulating the global model to produce targeted misclassifications (e.g., misclassifying a stop sign as a speed limit sign). The prevailing theory posited that because DP enforces strict gradient clipping and drowns small statistical anomalies in Gaussian noise, it would systematically neutralize the stealthy perturbations required to implant a backdoor⁸⁸.

Recent empirical evaluations have definitively shattered this assumption, revealing a fundamental paradox: **DP noise masks the statistical fingerprints of Byzantine updates, inadvertently shielding them from detection**⁸⁷.

State-of-the-art robust aggregation mechanisms on the server—such as Multi-Krum, FLTrust, and MESAS—rely on analyzing the Euclidean distance, cosine similarity, or frequency-domain components of incoming updates to isolate and discard outliers⁸⁸. In a non-private setting, poisoned updates form a distinct cluster that these defenses easily filter out. However, in a DP-FL setting, the benign updates are heavily distorted by high-variance Gaussian noise. Consequently, the statistical boundary between benign and malicious updates collapses, rendering the robust aggregators blind⁸⁹.

1.1 The RING Attack and Exploitation

Sophisticated adversaries actively weaponize this masking effect. The recently proposed "RING" attack explicitly exploits DP to conceal malicious contributions⁸⁷. Compromised clients coordinate to craft adversarial perturbations that perfectly mimic the statistical distribution of DP-induced noise⁸⁷. Drawing inspiration from secret sharing protocols, these perturbations conceal the true backdoor payload locally, allowing the updates to bypass server-side anomaly detection. Upon aggregation at the server, the carefully calibrated adversarial noise cancels out, reconstructing a highly potent backdoor signal in the global model⁸⁷. The RING attack achieves over 90% attack success rates against state-of-the-art defenses under moderate privacy budgets, an improvement of 26x over baseline strategies⁸⁷.

1.2 Resolving the Tension

Addressing this vulnerability requires unified frameworks that co-optimize DP and robustness.

Frameworks like Byz-Clip21-SGD2M utilize a "double momentum" technique paired with robust aggregation⁹⁰. By averaging the updates of each client over time, the algorithm reduces the variance of honest clients (smoothing out the DP noise) while accumulating and exposing the consistent directional bias of the Byzantine attackers, allowing for accurate filtering without violating DP bounds⁹⁰. Similarly, DGShield employs a dual-phase group-wise aggregation approach, grouping updates based on magnitude deviation and applying similarity filtering at the group level to bypass the interference of individual-level DP noise⁹².

2. Addressing Statistical Heterogeneity (Non-IID Data)

FL deployments inherently suffer from non-IID data distributions, where different clients possess vastly divergent class labels and feature representations⁹⁴. In standard FL, this heterogeneity induces client drift. The introduction of DP aggressively exacerbates this drift²⁸.

Because DP-FedAvg applies a universal clipping threshold C , clients whose data distributions lie far from the global mean produce larger gradients, which are subsequently clipped more heavily. This disproportionately suppresses the contributions of minority distributions, skewing the global model toward the majority class and degrading overall fairness and accuracy⁹⁷. To overcome this, DP-SCAFFOLD integrates DP constraints into the variance-reducing SCAFFOLD architecture³⁷. It introduces local and global control variates to correct the update direction of individual clients *before* clipping, ensuring that the clipped, noisy updates accurately guide the global model toward the true optimum despite high heterogeneity³⁷. Additionally, PrivateFL explicitly learns a differentially private, personalized data transformation for each client during local training, shifting the client's data distribution to explicitly compensate for the heterogeneity induced by DP clipping bias²⁸.

Conclusion

The integration of Differential Privacy into Federated Learning constitutes a critical evolution in the quest for secure, distributed artificial intelligence. By providing mathematically rigorous bounds on information leakage, DP-FL directly neutralizes devastating vulnerabilities such as gradient inversion and membership inference attacks. As detailed in this survey, the field is rapidly transitioning from rudimentary Central and Local DP models toward highly sophisticated Distributed and Shuffle architectures. Through the utilization of Discrete Gaussian mechanisms, cryptographic Secure Aggregation, and privacy amplification by shuffling, these modern frameworks achieve near-optimal privacy-utility trade-offs while operating in untrusted environments.

Simultaneously, the discipline has proven its adaptability by extending DP constraints to cross-silo Vertical FL architectures and the parameter-efficient fine-tuning of massive LLMs via LoRA. However, the path forward is complex. The profound tension between DP-induced noise and Byzantine robustness exposes a critical security gap, demonstrating that privacy mechanisms can be inadvertently weaponized to cloak malicious backdoors. Resolving this paradox, alongside mitigating the compounding effects of non-IID data heterogeneity, will dictate the trajectory of future research. Ultimately, the harmonious co-design of

cryptographic primitives, rigorous statistical privacy accounting, and robust optimization algorithms will be paramount to realizing the full potential of privacy-preserving machine learning.

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